Intro to AI Engineer: Prereq & Preparation

Super AI Engineer Season 6

Level 2 / 20 April 2026

Presented by

Aslan B. Wong

Arttie

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2 HCAI 101: THE THREE PILLARS

3 Data Scientist: BGC X Job Simulation

4 Review Math for ML

5 Model walkthrough

6 Gen AI: BGC X Job Simulation

01-R

INTERNATIONAL UNIVERSITY OF APPLIED SCIENCES

AB Wong

Ph.D. In Parallel Information Processing, Shenzhen University
ADVISOR: PROF. KAISHUN WU: IEEE FELLOW, HKUST(GZ): VICE-PRESIDENT (RESEARCH)

Aslan B. Wong (王泰华)
Other names
Researcher
Verified email at bu.edu
cyber-physical systems (C... iCALL articulation speech communication AIED

RESEARCH AREA

- An interdisciplinary research in AI-Human-Computer Interaction (AI-HCI) focused on developing innovations from
  - Interactive Intelligence: Human-AI Collaboration
  - Cyber-physical system: Smart Sensing
  - Cognitive Science to solve real-world problems.
  - Human-Data interactive visualization

RESEARCH PROFILE

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Google Scholar: Aslan B. Wong
ORCID ID: 0000-0003-4075-1485
ResearcherID: AAV-9326-2021

01-T

INTERNATIONAL UNIVERSITY OF APPLIED SCIENCES

AB Wong

Education
Ph.D. *Parallel Information Processing*, Shenzhen University
Master, Financial Management; Master, *AI for Business*, University Isabel I of Castile
M.Sc., *Management of Innovation and Emerging Technologies*
AA., *Interdisciplinary Studies*, Excelsior, Boston, Harvard University
BFA, *Beaux Art et Design*, Sorbonne

Work Experiences
Chair: ACM Chapter (Thailand)
AI Consultant – Thai-China Chamber of Commerce (Shanghai)
CTO – Internet & Application Service Firm, Shanghai
CEO – Articulation Sensing Startup, Shenzhen

AREA

- An interdisciplinary researcher focused on developing innovations from
  - Business Intelligence
  - M&A & Due diligence
  - Transformation

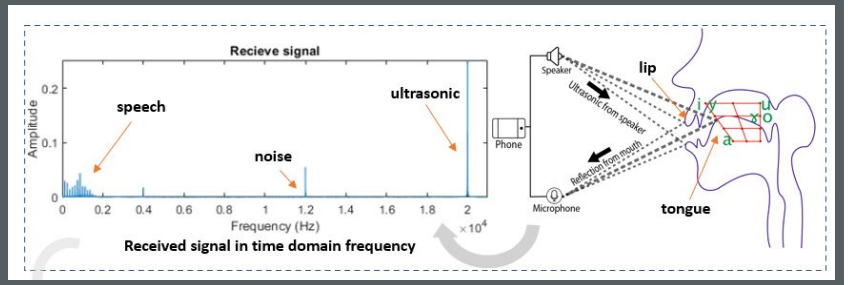
MORE SMALL WORKSHOPS

Articulation Sensing & Transparency

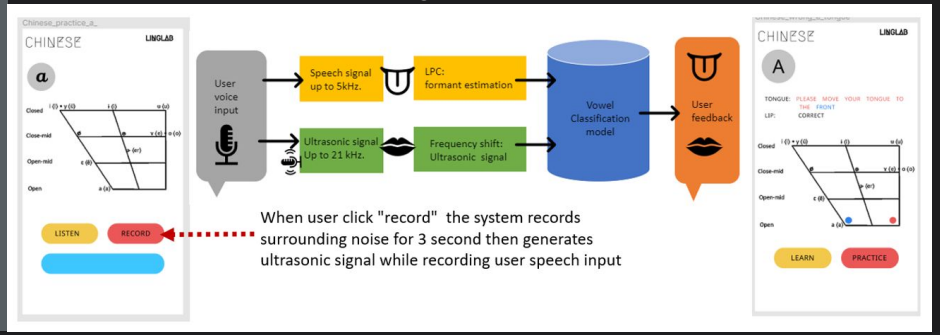
A Multifaceted Approach Advancing HCI Applications via Smartphones

1. Best award at the ACM UbiComp 2021
2. IEEE 19th Annual PerCom 2021
3. ACM 34th Annual UIST 2021
4. ACM 20th Annual MobiSys 2022
5. Speech Communication Q1 (96% #36/1126)

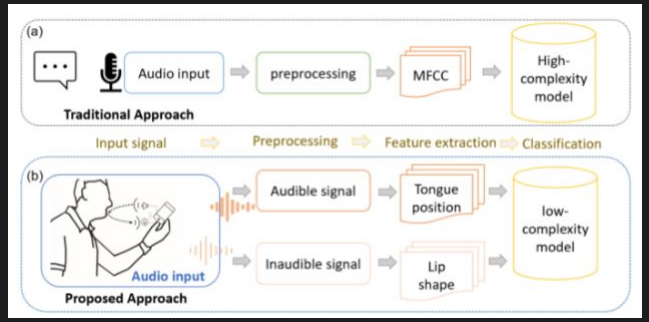
to sense articulation by speech and the **Doppler effect of the ultrasonic signal**



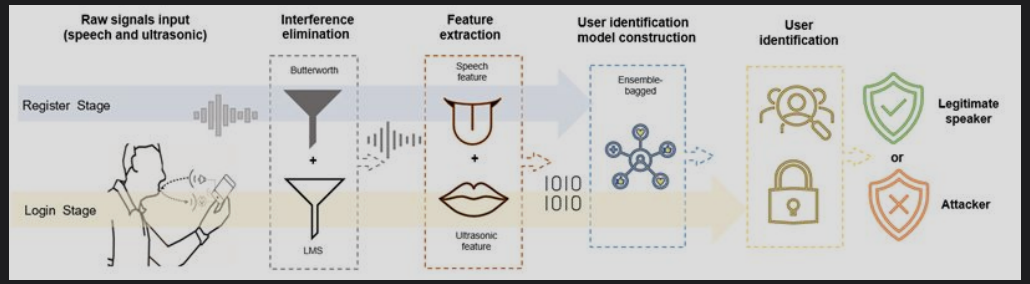
2. Pronunciation Training



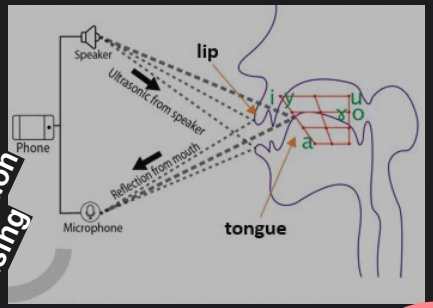
1. Articulation Verification

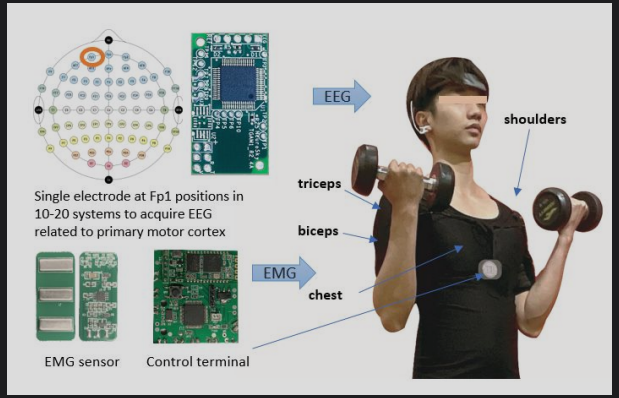


3. User Authentication



HUMAN-CENTRIC ARTIFICIAL INTELLIGENCE





Articulation Sensing

- **LingLab:** Articulation Detection and Analysis System
- **Artilock:** Smartphone Voice-articulation Authentication System
- **OsLing:** Pronunciation Training System Development

Healthcare AI Application

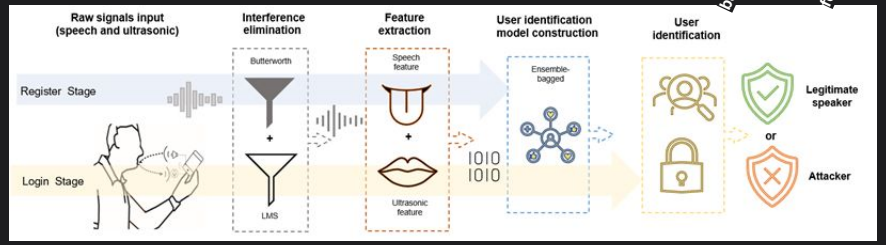
- Respiratory diagnosis of lung function in COPD patients using smartphones
- Measuring pulmonary function parameters based on speech signals
- Interpretable AI System for Pneumonia Detection

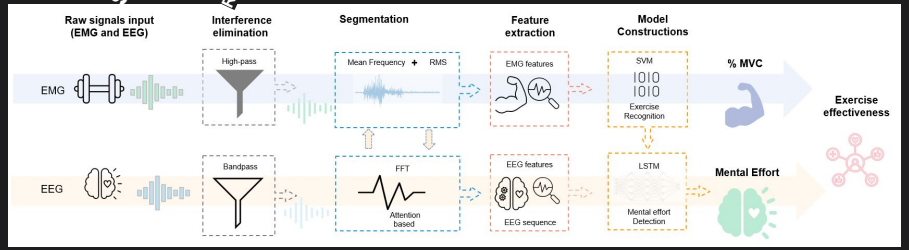
IoT assisted activities sensing & HCI

- Fa-Fall: Study on Full Range of Acoustic Signal for Fall Detection
- VitSense: Neuromuscular Activity Monitoring for Strength Training
- IoT for cognitive monitoring system

Neuroscience

- multi-channel tDCS approach targeting the ventromedial prefrontal cortex (VMPFC)

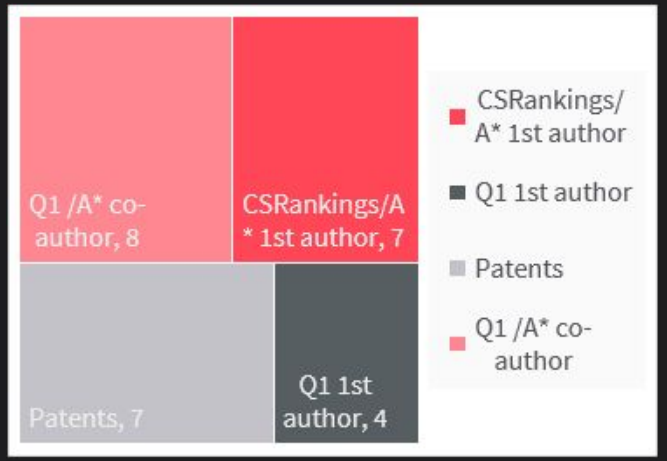


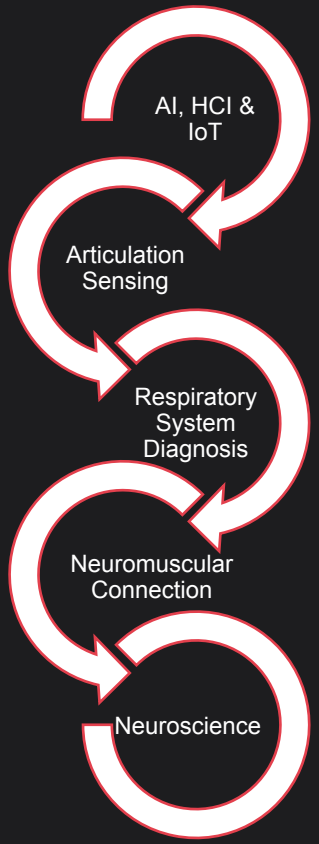


Research Experience

– Key Achievements:

- 11 articles as 1st Author : published in 4 Q1 journals and presented at 7 CSRankings/A* conferences.
- Inventor on 8 patents in AI and sensing technologies.
- Developed innovative systems for speech articulation sensing, human behavior monitoring, and interpretable AI for healthcare.





Research Project lead Experience

College of Computer Science & Software Engineering, Shenzhen University

Institute for Healthcare AI Application, Guangdong Second Provincial General Hospital

Shenzhen-Hong Kong Institute of Brain Science

Department of Psychology, University of Wisconsin-Milwaukee

Institute of Brain and Psychological Sciences, Sichuan Normal University

HCAI 101: THE THREE PILLARS

What is AI Hallucination ?

It is the phenomenon where AI creates information that appears credible, but is incorrect, inaccurate, or non-existent.

HCAI 101: THE THREE PILLARS
WE DESIGN FOR HUMAN COLLABORATION, NOT JUST AUTOMATION.

- 1. **Trust** (ความเชื่อถือ):
  A measure of human reliance (Social Science).

- 2. **Interpretability** (ความโปร่งใส):
  The model must explain its reasoning (Science/Engineering)

- 3. **Fairness** (ความเป็นธรรม):
  Decisions must be equitable across all user groups (Arts/Ethics).

- Can humans rely on the output?

- Can the model explain ***why*** it decided?

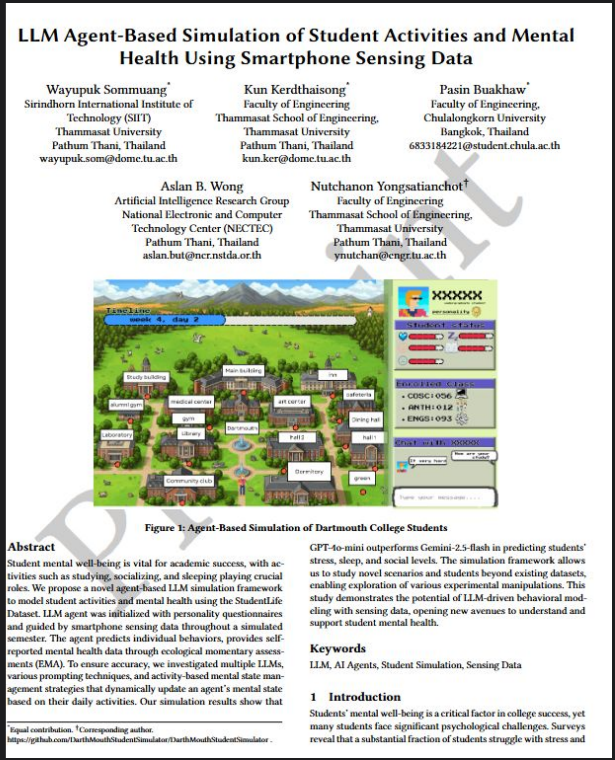
- Is the decision equitable?

PRIMARY CASE STUDY: LLM AGENTS FOR MENTAL HEALTH (FUSION)

- **Project: LLM Agent-Based Simulation of Student Activities and Mental Health.**

- **MAS Fusion:**
  - **Science/Tech**
  - **Arts/Social Science**

- **Takeaway:**



LLM Agent-Based Simulation of Student Activities and Mental Health Using Smartphone Sensing Data

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Figure 1: Agent-Based Simulation of Dartmouth College Students

Abstract

Student mental well-being is vital for academic success, with activities such as studying, socializing, and sleeping playing crucial roles. We propose a novel agent-based LLM simulation framework to model student activities and mental health using the StudentLife Dataset. LLM agent was initialized with personality questionnaires and guided by smartphone sensing data throughout a simulated semester. The agent predicts individual behaviors, provides self-reported mental health data through ecological momentary assessments (EMA). To ensure accuracy, we investigated multiple LLMs, various prompting techniques, and activity-based mental state management strategies that dynamically update an agent's mental state based on their daily activities. Our simulation results show that GPT-4o-mini outperforms Gemini-2.5-flash in predicting students' stress, sleep, and social levels. The simulation framework allows us to study novel scenarios and students beyond existing datasets, enabling exploration of various experimental manipulations. This study demonstrates the potential of LLM-driven behavioral modeling with sensing data, opening new avenues to understand and support student mental health.

Keywords

LLM, AI Agents, Student Simulation, Sensing Data

1 Introduction

Students' mental well-being is a critical factor in college success, yet many students face significant psychological challenges. Surveys reveal that a substantial fraction of students struggle with stress and

*Equal contribution. †Corresponding author.
https://github.com/DarthMouth/StudentSimulator/DarthMouthStudentSimulator .

arxiv.org/pdf/2508.02679

PRIMARY CASE STUDY: LLM AGENTS FOR MENTAL HEALTH (FUSION)

- **Project: LLM Agent-Based Simulation of Student Activities and Mental Health**.

- **MAS Fusion:**
  - **Science/Tech**
  - **Arts/Social Science**

- **Takeaway:**

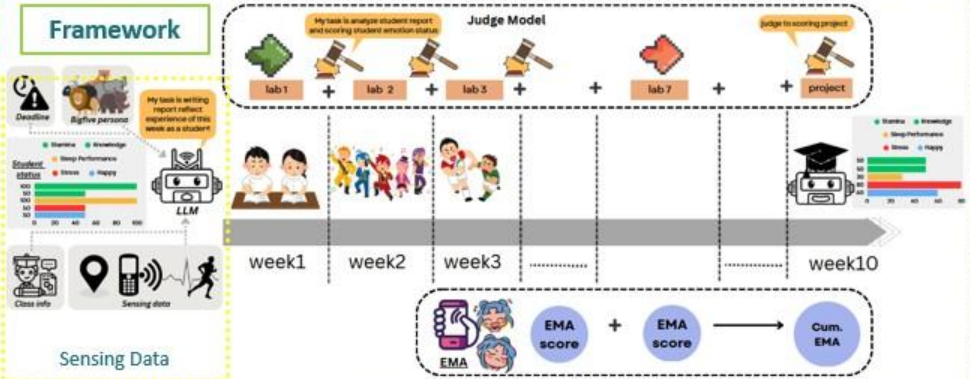
- **Abstract:**

Student mental well-being is vital for academic success, with activities such as studying, socializing, and sleeping playing crucial roles. We propose a novel agent-based LLM simulation framework to model student activities and mental health using the StudentLife Dataset. LLM agent was initialized with personality questionnaires and guided by smartphone sensing data throughout a simulated semester. The agent predicts individual behaviors, provides self reported mental health data through ecological momentary assessments (EMA). To ensure accuracy, we investigated multiple LLMs, various prompting techniques, and activity-based mental state management strategies that dynamically update an agent's mental state based on their daily activities. Our simulation results show that GPT-4o-mini outperforms Gemini-2.5-flash in predicting students' stress, sleep, and social levels. The simulation framework allows us to study novel scenarios and students beyond existing datasets, enabling exploration of various experimental manipulations. This study demonstrates the potential of LLM-driven behavioral modeling with sensing data, opening new avenues to understand and support student mental health.

arxiv.org/pdf/2508.02679 LLM Agent-Based Simulation of Student Activities and Mental Health Using Smartphone Sensing Data

The **framework** is specifically designed to simulate the complex interplay between students' behaviors and their psychological states throughout a semester

2 Framework

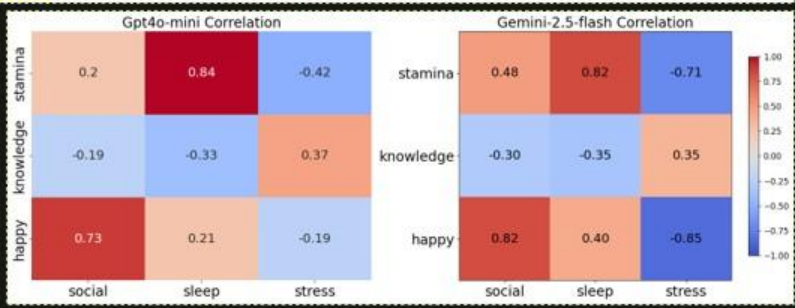


3 Evaluations

Stress Level:
GPT-4o-mini (MAE: 0.675) > Gemini-2.5-flash (MAE:0.750).

Sleep Level:
GPT-4o-mini (MAE: 0.963) > Gemini-2.5-flash (MAE:1.245).

Social Level:
GPT-4o-mini (MAE: 0.250) > Gemini-2.5-flash (MAE: 0.282)



4

1. The simulation allows researchers to **explore "what if" scenarios** by safely testing interventions
2. Unlike traditional models, these AI agents can be **"interviewed,"** allowing researchers to ask them hypothetical questions to gather rich, qualitative insights
3. Ultimately, this project demonstrates **the immense promise of combining AI with real-world** data, creating powerful and unprecedented avenues

the **objective** of the LLM Agent-Based Simulation framework is to model student activities and mental well-being to better understand and support mental health.

1

The Study aims to shift from observing student data to interacting with it, creating a risk-free 'digital twin' of the campus ecosystem to proactive well-being.

LLM Agent Simulation of Student Activities and Mental Health Using Smartphone Sensing Data

Source code

Demo-video

UBICOMP ISWC2025

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Sirindhorn International Institute of Technology (SIIT) ; Thammasat School of Engineering, Thammasat University; Faculty of Engineering Chulalongkorn University
Artificial Intelligence Research Group, National Electronic and Computer Technology Center (NECTEC)

NECTEC NSTDA AiAT

PRIMARY & SECONDARY CASE STUDY: FUSION + **HCAI**

- **Project: LLM Agent-Based Simulation of Student Activities and Mental Health**.
- **MAS Fusion:**
  - **Science/Tech** (LLM Agents, Smartphone Sensing Data) predicts behaviors;
  - **Arts/Social Science** (Personality, EMAs, Mental Health) provides the human context.
- **Takeaway:** A blueprint for fusing qualitative and quantitative data in MAS research.

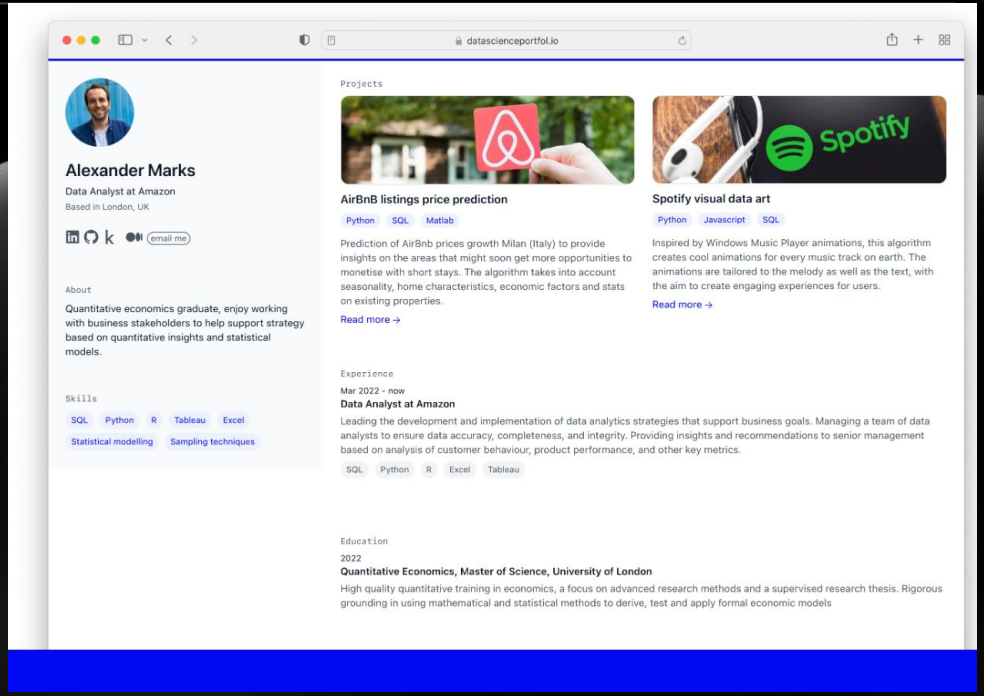
- **Projects:** Speech Articulation Sensing (Artilock/OsLing) and Interpretable AI for Pneumonia.
- **Connection to HCAI:**
  - Even non-mental health sensing must build **Transparency**.
  - Show the diagram of using **ultrasonic signals** to sense **tongue and lip movement**.
- **Takeaway:** Scientific rigor must serve human trust.

arxiv.org/pdf/2508.02679

HCAI is the lens through which we structure meaningful interdisciplinary research

Portfolio Preparation

Build and share your data portfolio.

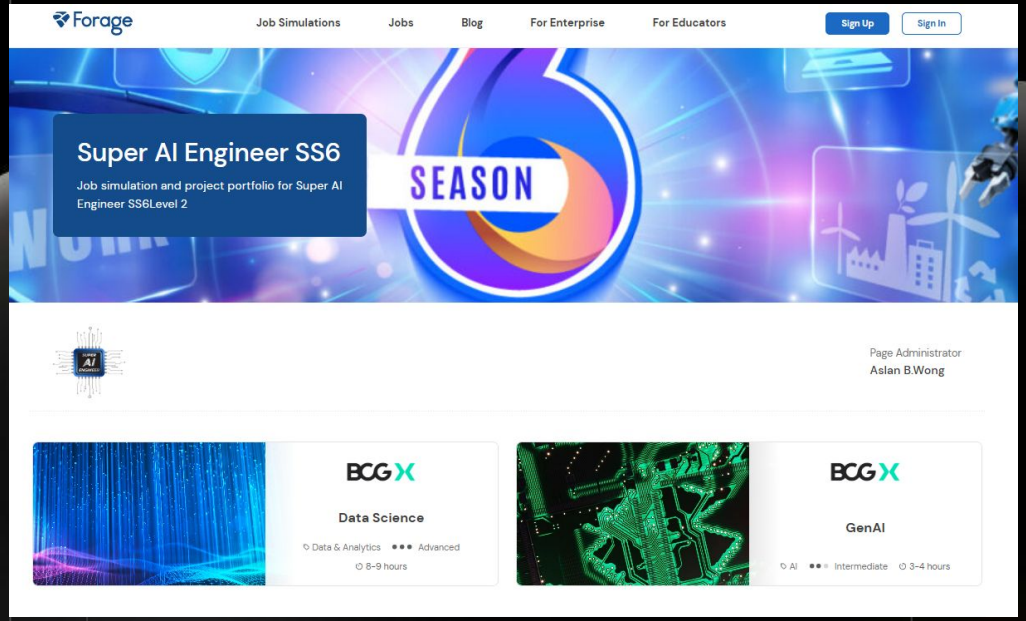


Scan me!

Everything you need to create high quality data projects and showcase them on your own portfolio website.
Link: https://www.datascienceportfol.io/

Data Scientist: BGC X Job Simulation

Data Scientist: BGC X Job Simulation

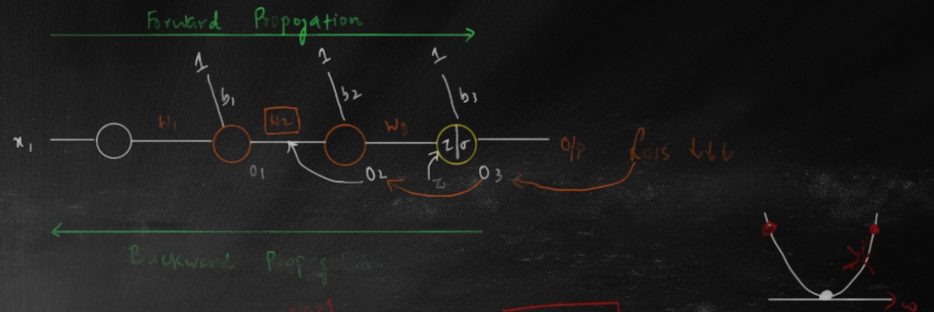


Click: Job simulation and project portfolio for Super AI Engineer SS6 **Level 2**

Scan me!

Review Math for ML

Our Main mission today!



Please Grab your pencil & papers!

Review Math for ML: Gradient Descent Algorithm

Doing by Hand understand by Heart.

Review Math for ML

1

Linear Regression

2

Cost Function

3

Gradient Descent

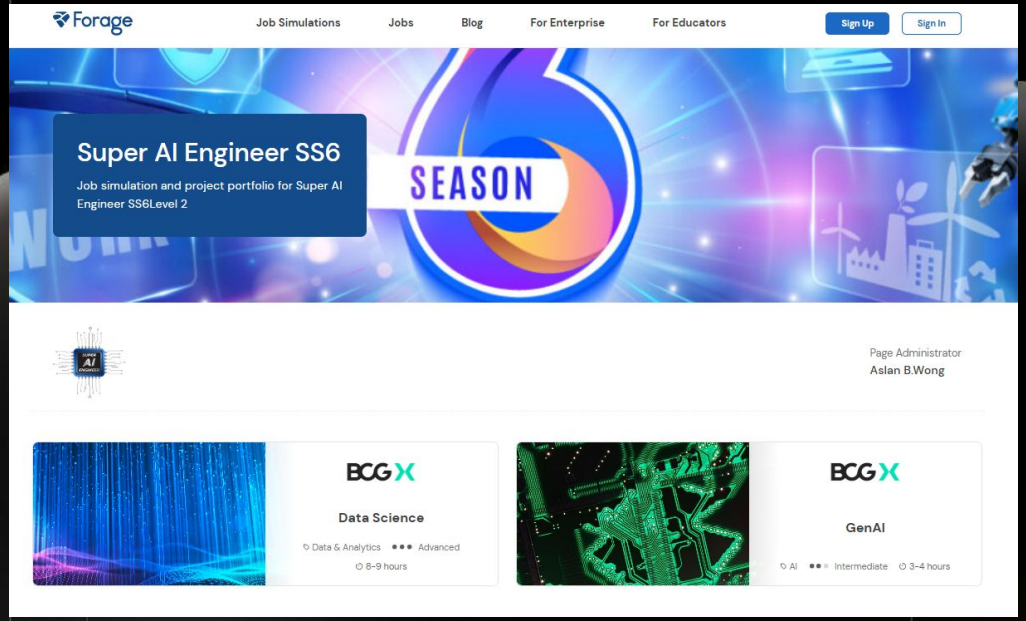
ML model walkthrough

LINEAR REGRESSION REVIEW

Let' go to Google Collab!

Gen AI: BGC X Job Simulation

Gen AI: BGC X Job Simulation



Scan me!

Job simulation and project portfolio for Super AI Engineer SS6 **Level 2**

Thank you

Ready for what's next?

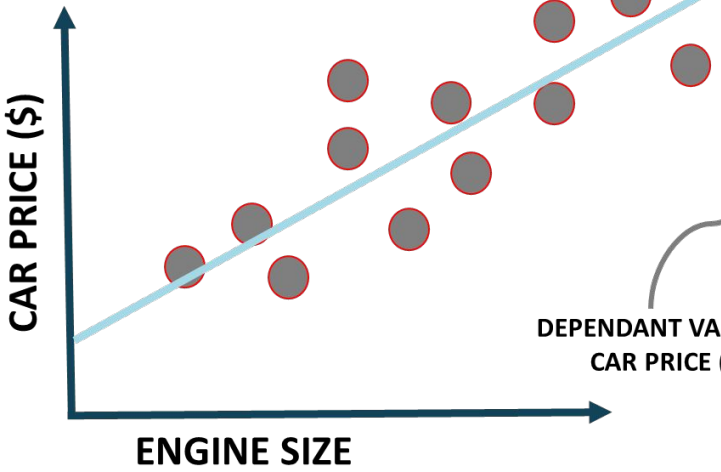
Let's talk

Appendix

Review & note

Appendix

SIMPLE LINEAR REGRESSION



MODEL! (GOAL)

$$y = b + m * x$$

DEPENDANT VARIABLE
CAR PRICE ($)

INDEPENDENT VARIABLE
ENGINE SIZE

Goal is to obtain a relationship (model) between two variables only such as age and insurance cost for example.